7. It is somewhat of a misnomer to speak of the "center" of a black hole as if it were a place in space. The reason, roughly speaking, is that when one crosses a black hole's event horizon—its outer edge—the roles of space and time are interchanged. In fact, just as you can't resist going from one second to the next in time, so you can't resist being pulled to the black hole's "center" once you've crossed the event horizon. It turns out that this analogy between heading forward in time and heading toward a black hole's center is strongly motivated by the mathematical description of black holes. Thus, rather than thinking of the black hole's center as a location in space, it is better to think of it as a location in time. Furthermore, since you can't go beyond the black hole's center, you might be tempted to think of it as a location in spacetime where time comes to an end. This may well be true. But since the standard general relativity equations break down under such extremes of huge mass density, our ability to make definite statements of this sort is compromised. Clearly, this suggests that if we had equations that don't break down deep inside a black hole, we might gain important insights into the nature of time. That is one of the goals of superstring theory.

8. As in earlier chapters, by "observable universe" I mean that part of the universe with which we could have had, at least in principle, communication during the time since the bang. In a universe that is infinite in spatial extent, as discussed in Chapter 8, all of space does *not* shrink to a point at the moment of the bang. Certainly, everything in the observable part of the universe will be squeezed into an ever smaller space as we head back to the beginning, but, although hard to picture, there are things—infinitely far away—that will forever remain separate from us, even as the density of matter and energy grows ever higher.

9. Leonard Susskind, in "The Elegant Universe," *NOVA*, three-hour PBS series first aired October 28 and November 4, 2003.

10. Indeed, the difficulty of designing experimental tests for superstring theory has been a crucial stumbling block, one that has substantially hindered the theory's acceptance. However, as we will see in later chapters, there has been much progress in this direction; string theorists have high hopes that upcoming accelerator and space-based experiments will provide at least circumstantial evidence in support of the theory, and with luck, maybe even more.